MOHAMMED IRFAN KURPATH

PhD Student in Computer Vision
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Google Scholar | **GitHub** | **LinkedIn**

PROFILE

I'm a PhD student in Computer Vision at MBZUAI in Abu Dhabi. My research centers on building multimodal AI systems for **proactive, personalized assistance**: systems that can see, listen, remember, reason, and speak, for useful assistance in day-to-day life.

Interests: omni-modal LLMs, proactive agents, omni-modal embedding and retrieval, long-term multimodal memory, real-time speech-to-speech, multilingual MLLMs, assistive technology.

RESEARCH HIGHLIGHTS

- First and co-author publications at leading AI venues including **NeurIPS, EMNLP, and ACL**, with contributions across omni-modal long-video reasoning, omni-modal embeddings, streaming speech generation, and multimodal medical AI.
- **1st place** in the EgoConv track and **runner-up** in the EgoLongQA track of the **Wearable AI Grand Challenge @ ECCV 2026**, with entries building on LongShOT.
- **Meta Llama Impact Innovation Award** for **BiMediX2**, an Arabic-English medical multimodal model, accepted to **EMNLP 2025 Findings** and **featured on LinkedIn by Yann LeCun**.
- Co-author on **MAviS**, recognized with the **EMNLP 2025 Senior Area Chair Highlight Award**.
- Research supported by the **NVIDIA Academic Grant** and the **MBZUAI–IIT Delhi Research Grant**, with work showcased at GITEX 2024 and 2025, Machines Can Think 2026, and the AI4SD side event of the 79th UN General Assembly.
- Strong academic and open-source impact, with **100+ citations**, **600+ GitHub stars**, and **55K+ downloads** of publicly released models and datasets.

PUBLICATIONS (* equal contribution)

LongShOT: A Benchmark for Omni-Modal Reasoning in Long Videos
Mohammed Irfan Kurpath*, *Jaseel Muhammad Kaithakkodan**, *Jinxing Zhou, Sahal Shaji Mullappilly, Mohammad Almansoori, Noor Ahsan, Beknur Kalmakhanbet, Sambal Shikhar, Rishabh Lalla, Jean Lahoud, Mariette Awad, Fahad Shahbaz Khan, Salman Khan, Rao Muhammad Anwer, Hisham Cholakkal*
NeurIPS 2026 | Project Page | arXiv | GitHub | HF Dataset

Omni-Embed-Mini: Binding Modalities Without Forgetting via Dense Distillation
Mohammed Irfan Kurpath*, *Jaseel Muhammad Kaithakkodan, Sahal Shaji Mullappilly, Ivan Laptev, Hisham Cholakkal*
EMNLP 2026 – Findings | Project Page | GitHub | HF Collection

MediX-R1: Open Ended Medical Reinforcement Learning
*Sahal Shaji Mullappilly**, ***Mohammed Irfan Kurpath****, *Jaseel Muhammad Kaithakkodan, Omair Mohamed, Mohamed Zidan, Fahad Khan, Salman Khan, Rao Anwer, Hisham Cholakkal*
EMNLP 2026 – Findings | Project Page | arXiv | GitHub | 36K+ HF Downloads
8B model outperforms the 27B MedGemma; 2B variant runs on-device on a phone.

Hard Vision, Easy Vision: What GPT-6 Astra Reveals Across Computer Vision
*Hanoona Rasheed**, ***Mohammed Irfan Kurpath****, *Bin Ren, Hisham Cholakkal, Fahad Shahbaz Khan, Salman Khan*
arXiv 2026 | Project Page | GitHub

BiMediX2: Bio-Medical EXpert LMM for Diverse Medical Modalities
*Sahal Shaji Mullappilly**, ***Mohammed Irfan Kurpath****, *Sara Pieri, Saeed Yahya Alseiari, Shanavas Cholakkal, Khaled M Aldahmani, Fahad Shahbaz Khan, Rao Muhammad Anwer, Salman Khan, Timothy Baldwin, Hisham Cholakkal*
EMNLP 2025 – Findings | Meta Llama Impact Award | LinkedIn Feature | GitHub | 10K+ HF Downloads

LLMVoX: Autoregressive Streaming Text-to-Speech Model for Any LLM
*Sambal Shikhar, **Mohammed Irfan Kurpath**, Sahal Shaji Mullappilly, Jean Lahoud, Fahad Shahbaz Khan, Rao Muhammad Anwer, Salman Khan, Hisham Cholakkal*
ACL 2025 – Findings | Project Page | GitHub (300+ stars)

MAviS: A Multimodal Conversational Assistant For Avian Species
*Yevheniia Kryklyvets, **Mohammed Irfan Kurpath**, Sahal Shaji Mullappilly, Jinxing Zhou, Fahad Shahbaz Khan, Rao Muhammad Anwer, Salman Khan, Hisham Cholakkal*
EMNLP 2025 | SAC Highlight Award | GitHub

A Culturally-Diverse Multilingual Multimodal Video Benchmark & Model
*Bhuiyan Sanjid Shafique, Ashmal Vayani, Muhammad Maaz, Hanoona Abdul Rasheed, Dinura Dissanayake, **Mohammed Irfan Kurpath**, et al., Salman Khan, Fahad Shahbaz Khan*
EMNLP 2025 | Project Page | HF Benchmark

An IMUs and Potentiometer-Based Controller for Robotic Arm-Hand Teleoperation
Mohammed Irfan Kurpath*, *Pradeep Kumar Adwai, Janardhan Bodireddy, Chandrasekaran K, Navin Sam K*
Sensors and Actuators A: Physical, 2024 | DOI

EXPERIENCE

Aug 2025 to Present — **PhD Student, Computer Vision** · MBZUAI, Abu Dhabi (advised by Dr. Hisham Cholakkal)
Focus: multimodal AI systems for personalized assistance: omni-modal embeddings for memory and retrieval, evidence-seeking agents for long-video understanding, and multimodal models for healthcare.
Responsibilities: lead projects end to end, from problem formulation and benchmark design to model training on multi-node GPU clusters, evaluation and paper writing; mentor students and review for CVPR and NeurIPS.

Mar 2024 to Aug 2025 — **Research Associate** · MBZUAI, Abu Dhabi
Focus: medical and speech-enabled multimodal LLMs, and multilingual multimodal evaluation.
Responsibilities: co-led a bilingual Arabic-English medical LMM, contributed to streaming speech generation for LLMs and to the evaluation toolkit of a multilingual video benchmark; built and presented live demos at GITEX, UNGA79, Abu Dhabi Sustainability Week and Machines Can Think; mentored an MSc thesis and UGRIP interns.

Jul 2022 to Feb 2024 — **Project Associate** · CEET, IIT Madras Research Park, Chennai
Focus: multilingual speech and sign-language technology for Indian languages.
Responsibilities: built voice-command banking across seven Indian languages (ASR, MT, TTS and transliteration), Indian Sign Language recognition from 3D pose, and a speech-controlled wheelchair guide with a real-time chatbot; mentored summer interns.

AWARDS & GRANTS

2026 — **Wearable AI Grand Challenge @ ECCV 2026**: **1st place**, EgoConv track (StreamShot: streaming multi-turn conversation over long egocentric video) and **runner-up**, EgoLongQA track (LongShotV2: long egocentric video QA). Entries build on LongShOT (first author, NeurIPS 2026).

2025 — **NVIDIA Academic Grant**: to advance clinical reasoning capabilities of healthcare LLMs and LMMs. PI: Dr. Hisham Cholakkal. Role: co-lead student.

2025 — **Meta Regional Research Grant**: for Project OMER, building multimodal LLMs for smart wearables. PI: Dr. Hisham Cholakkal. Role: lead student.

2024 — **Meta Llama Impact Innovation Award**: for the project BiMediX2 (EMNLP 2025 Findings), building an Arabic-English bilingual medical LMM. PI: Dr. Hisham Cholakkal. Role: co-lead student.

2021 — **Bachelor of Technology**: First Class with Distinction.

DEMOS & EXHIBITIONS

- **Personalized wearable assistants for elderly care** with smartwatches, smart glasses, and non-invasive EEG devices, at **GITEX 2025** and **Machines Can Think (MCT) 2026**.
- **Multilingual multimodal LLM integrated with a robotic dog** (Unitree GO2) that holds **interactive conversations** and **carries out requested physical actions** while conversing, at **GITEX 2024** and **Abu Dhabi Sustainability Week 2025**.
- **BiMediX2 assistant** deployed over **Telegram**, presented at the **AI4SD event at UNGA79** and **GITEX 2024**. [Demo]

EDUCATION

2025 to Present — **PhD, Computer Vision**, MBZUAI — GPA: 3.77/4

2017 to 2021 — **B.Tech, Electrical and Electronics Engineering**, NIT Puducherry — CGPA: 8.66/10

SERVICE & MENTORING

- **Reviewer**: CVPR, NeurIPS, and Springer Nature journals.
- **Mentor**, **AI for Africa Bootcamp 2026** (MBZUAI × African Leadership Academy, Johannesburg): graduate mentor to high school students.
- **Mentor**, MBZUAI: MSc thesis (MAviS, EMNLP 2025) and UGRIP undergraduate interns.
- **Mentor**, CEET-IIT Madras Research Park: summer interns.

SKILLS

Languages & Tools	Python, Git, Linux
Deep Learning	PyTorch, Hugging Face Transformers, OpenCV, vLLM
Training & Infra	LLM and multimodal LLM training and fine-tuning, distributed and multi-node training, SLURM, AWS HyperPod-class GPU clusters, NVIDIA and AMD clusters
Speech	ASR, TTS, real-time speech-to-speech pipelines
Robotics & Edge	ROS, Unitree GO2 quadruped, NVIDIA Jetson, Raspberry Pi, Arduino, ESP32, embedded sensors, IoT

OTHER PROJECTS

Oct 2024 — **Real-time Multimodal Conversational Model**: speech-in, speech-out over a lightweight VLM, integrated into the **Unitree GO2 robot dog** and later extended into the **personalized wearable assistants for elderly care** demonstrated at GITEX 2025 and MCT 2026. [GITEX]

Dec 2022 — **Fun With Image Processing (Build Lab Course)**: an introductory, hands-on OpenCV course for beginners from diverse backgrounds, offered by the Build Club at IITMRP with partner colleges. [GitHub]